

Ryan Kazuo Cosner

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RESEARCH INTERESTS

I leverage tools from machine learning and control theory to build safe and performant robotic systems that we can trust. I am particularly interested in navigating real-world perception-based uncertainty and tackling complex, data-sparse learning problems in safety-critical robotics.

APPOINTMENTS

Tufts University

Glenn R. Stevens Assistant Professor, Mechanical Engineering

2026–

Visiting Assistant Professor, Mechanical Engineering

2025–2026

EDUCATION

California Institute of Technology

PhD, Mechanical Engineering

2019–2025

MS, Mechanical Engineering

2019–2021

University of California, Berkeley

BS, Mechanical Engineering

2015–2019

PAST POSITIONS

Nvidia Corporation

Research Intern, Advisor: Prof. Marco Pavone

Summer 2022

Squishy Robotics

Mechatronics Intern, Advisor: Prof. Alice Agogino

Summer 2019

Ford Motor Company

Vehicle Evaluation and Verification Intern

Summer 2018

ACADEMIC PUBLICATIONS

** indicates equal contribution*

Journal Articles

- [1] **Ryan K. Cosner**, Preston Culbertson, Aaron D. Ames. *Bounding Stochastic Safety: Leveraging Freedman's Inequality with Discrete-Time Control Barrier Functions*. IEEE, Control Systems Letters (CSL), 2024. ([pdf](#))
- [2] Max H. Cohen, **Ryan K. Cosner**, Aaron D. Ames. *Constructive Safety-Critical Control: Synthesizing Control Barrier Functions for Partially Feedback Linearizable Systems*. IEEE Control Systems Letters, 2024. ([pdf](#))

- [3] Devansh R. Agrawal*, Hardik Parwana*, **Ryan K Cosner***, Ugo Rosolia, Aaron D. Ames, Dimitra Panagou. *A Constructive Method for Designing Safe Multirate Control for Differentially-Flat Systems*. IEEE Control Systems Letters (CSL), 2021. ([pdf](#))
- [4] Tamas G. Molnar, **Ryan K. Cosner**, Andrew W. Singletary, Wyatt Ubellacker, Aaron D. Ames. *Model-Free Safety-Critical Control for Robotic Systems*. IEEE Robotics and Automation Letters (RAL), 2021. ([pdf](#))
- [5] Kunal Garg, **Ryan K. Cosner**, Ugo Rosolia, Aaron D. Ames, Dimitra Panagou. *Multi-rate Control Design under Input Constraints via Fixed-Time Barrier Functions*. IEEE Control Systems Letters (CSL), 2021. ([pdf](#))

Conference Papers

- [1] Alexandre Capone, **Ryan K. Cosner**, Aaron D. Ames, Sandra Hirche. *Learning Safe Control via On-the-Fly Bandit Exploration*. International Conference on Machine Learning (ICML), 2025. ([pdf](#))
- [2] Lizhi Yang*, Blake Werner*, **Ryan K. Cosner**, David Fridoviech-Keil, Preston Culbertson, Aaron D. Ames. *SHIELD: Safety on Humanoids via CBFs in Expectation on Learned Dynamics*. International Conference on Robotics and Control (ICRA), 2025. ([pdf](#))
- [3] Gilbert Bahati, **Ryan K. Cosner**, Max H. Cohen, Ryan M. Bena, Aaron D. Ames. *Control Barrier Function Synthesis for Nonlinear Systems with Dual Relative Degree*. Conference on Decision and Control (CDC), 2025. ([pdf](#))
- [4] **Ryan K. Cosner**, Igor Sadalski, Jana K. Woo, Preston Culbertson, Aaron D. Ames. *Generative Modeling of Residuals for Real-Time Risk-Sensitive Safety with Discrete-Time Control Barrier Functions*. International Conference on Robotics and Automation (ICRA), 2024. ([pdf](#))
- [5] Preston Culbertson, **Ryan K. Cosner**, Maegan Tucker, Aaron D. Ames. *Input-to-State Stability in Probability*. Conference on Decision and Control (CDC), 2024. ([pdf](#))
- [6] **Ryan K. Cosner**, Preston Culbertson, Andrew J. Taylor, Aaron D. Ames. *Robust Safety under Stochastic Uncertainty with Discrete-Time Control Barrier Functions*. Robotics: Science and Systems (RSS), 2023. ([pdf](#))
- [7] **Ryan K. Cosner**, Yuxiao Chen, Karen Leung, Marco Pavone. *Learning Responsibility Allocations for Safe Human-Robot Interaction with Applications to Autonomous Driving*. International Conference on Robotics and Automation (ICRA), 2023. ([pdf](#))
- [8] Shushant Veer, Karen Leung, **Ryan K. Cosner**, Yuxiao Chen, Marco Pavone. *Receding Horizon Planning with Rule Hierarchies for Autonomous Vehicles*. International Conference on Robotics and Automation (ICRA), 2023. ([pdf](#))
- [9] **Ryan K. Cosner**, Yisong Yue, Aaron D. Ames. *End-to-End Imitation Learning with Safety Guarantees using Control Barrier Functions*. Conference on Decision and Control (CDC), 2022. ([pdf](#))
- [10] Andrew J. Taylor*, Victor D. Dorobantu*, **Ryan K. Cosner***, Yisong Yue, Aaron D. Ames. *Safety of Sampled-Data Systems with Control Barrier Functions via Approximate Discrete Time Models*. Conference on Decision and Control (CDC), 2022. ([pdf](#))
- [11] **Ryan K. Cosner**, Maegan Tucker, Andrew J. Taylor, Kejun Li, Tamas G. Molnar, Anil Alan, Gabor Orosz, Yisong Yue, Aaron D. Ames. *Safety-Aware Preference-Based Learning for Safety-Critical Control*. Learning for Dynamics and Control Conference (L4DC), 2022. ([pdf](#))

- [12] **Ryan K. Cosner***, Ivan D. Jimenez Rodriguez*, Tamas G. Molnar, Wyatt Ubellacker, Yisong Yue, Aaron D. Ames, Katherine L. Bouman. *Self-Supervised Online Learning for Safety-Critical Control using Stereo Vision*. International Conference on Robotics and Automation (ICRA), 2022. ([pdf](#))
- [13] **Ryan K. Cosner**, Andrew W. Singletary, Andrew J. Taylor, Tamas G. Molnar, Aaron D. Ames. *Measurement-Robust Control Barrier Functions: Certainty in Safety with Uncertainty in State*. International Conference on Intelligent Robots and Systems (IROS), 2021. ([pdf](#))
- [14] Noel Csomay-Shanklin*, **Ryan K. Cosner***, Min Dai*, Andrew J. Taylor, Aaron D. Ames. *Episodic Learning for Safe Bipedal Locomotion with Control Barrier Functions and Projection-to-State Safety*. Learning for Dynamics and Control Conference (L4DC), 2021. ([pdf](#))
- [15] Sarah Dean, Andrew J. Taylor, **Ryan K. Cosner**, Benjamin Recht, and Aaron D. Ames. *Guaranteeing Safety of Learned Perception Modules via Measurement-Robust Control Barrier Functions*. Conference on Robotic Learning (CoRL), 2020. ([pdf](#))
Best Student Paper Finalist.

Patents

- [1] **Ryan K. Cosner**, Yuxiao Chen, Karen Leung, Marco Pavone. *Allocating responsibility for autonomous and semi-autonomous machine interactions and applications*. Pub. No.: US 2024/0160913 A1, 2024.
- [2] Shushant Veer, Karen Leung, **Ryan K. Cosner**, Yuxiao Chen, Marco Pavone. *Ego trajectory planning with rule hierarchies for autonomous vehicles*. Pub. No.: US 2024/0199074 A1. 2024.

Workshops and Invited Presentations

- [1] **Ryan K. Cosner**. *Stochastic Safety Guarantees for Real-World Systems using Discrete-Time Control Barrier Functions*. Southern California Controls Workshop, 2024.
- [2] **Ryan K. Cosner**. *CBF-based Safety for Real-World Robotic Systems*. Conference on Decision and Control (CDC) workshop, “Control Barrier Functions: Recent developments and future directions”, 2023.
- [3] Glen Chou, Shushant Veer, Heng Yang, **Ryan K. Cosner**, Marco Pavone. *Workshop on Safe and Robust Learning for Perception-based Control*. American Controls Conference (ACC), 2023. ([website](#))

TEACHING

2024: Teaching Assistant for Mobile Robotics: Applications, Prof. Günter Niemeyer

2024: Teaching Assistant for Mobile Robotics: Theory & Algorithms, Prof. Günter Niemeyer

2023: Teaching Assistant for Nonlinear Control, Prof. Aaron D. Ames

2023: Teaching Assistant for Nonlinear Dynamics, Prof. Aaron D. Ames

STUDENT ADVISING

1. Gilbert Bahati (Mechanical Engineering PhD student, 2024)
2. Igor Sadalski (Caltech Summer Undergraduate Research Fellow, 2023 – now at Somite Therapeutics)
3. Jana K Woo (Caltech Summer Undergraduate Research Fellow, 2023)

SERVICE AND OUTREACH

1. **Academic Peer Reviewer:** ACC, L4DC, CDC, MECC, NeuRips, ICRA, IROS, CDC, TAC, Automatica, L-CSS, RAL, TCST
2. **Caltech Center for Teaching, Learning, and Outreach**, Tour Guide. Provided lab tours for middle and high school students. 2021-Present.
3. **Caltech Triathlon**, President. Organized 3 weekly workouts and 15 team races, 2021-2024.
4. **Manhattan Beach Unified School District Community Panel for Equity**, Organizer. 2020.
5. **Science Olympiad**, Test Writer and Event Supervisor. Wrote exams and supervised events for the Southern California State-Level Science Olympiad Competition, 2023.
6. **Caltech Rise**, Volunteer Tutor. Tutored public school students struggling in math and science, 2019-2021.
7. **UC Berkeley**, Campus Ambassador. Gave tours of the UC Berkeley, 2016-2019.
8. **United Technologies for Kids**, Volunteer Teacher. Taught engineering to middle and high school students in English and Spanish in Peru, Summer 2017.

AWARDS

1. **Outstanding Teaching Assistant Award**, *Mechanical and Civil Engineering Department, Caltech*, 2024.
2. **Best Student Paper Finalist**, *Conference on Robotic Learning*, 2020.
3. **Graduate Student Fellowship**, *Rose Hills Foundation*, 2020.
4. **High Honors at Graduation**, *UC Berkeley*, 2019.
5. **Alexander and Ethel Levens Mechanical Engineering Award**, *UC Berkeley*, 2017.
6. **Regents' and Chancellor's Scholar**, *UC Berkeley*, 2015–2019.
7. **Chevron Academic Scholar**, *Chevron, El Segundo*, 2015.